

Unit 3 Pre-Test

Solving Linear Equations

Standard assessed: NC.8.EE.7

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.EE.7 Solve linear equations in one variable.

• *I can solve one-step and two-step equations using inverse operations.*

Match each vocabulary term to its definition, then solve each equation.

- v coefficient	A. Terms with the same variable raised to the same power.
- v constant	B. The value of the variable that makes the equation true.
- v inverse operations	C. A number multiplied by a variable.
- v solution	D. Operations that undo each other, such as + and −.
- v distributive property	E. A term with no variable — just a number.
- v like terms	F. Multiplying a factor across each term inside parentheses.

Keep the equation balanced — worked example

$$2x + 5 = 13 \rightarrow \text{subtract } 5 \rightarrow 2x = 8 \rightarrow \text{divide by } 2 \rightarrow \mathbf{x = 4}$$

Whatever you do to one side, do to the other.

1. Solve

$$x + 7 = 12$$

x =

2. Solve

$$x - 4 = 9$$

x =

3. Solve

$$3x = 21$$

x =

4. Solve

$$x \div 5 = 6$$

x =

5. Solve

$$2x + 3 = 11$$

x =

6. Solve

$$4x + 5 = 25$$

x =

SCORE 4.0 Target Content

NC.8.EE.7 Solve equations with the distributive property and variables on both sides.

• *I can solve multi-step equations, including variables on both sides.*

Solve each equation. Show your work.

7. Solve

$$2(x + 3) = 14$$

x =

8. Solve

$$3(x - 2) = 9$$

x =

9. Solve

$$5x + 2 = 3x + 10$$

x =

10. Solve

$$7x - 4 = 2x + 11$$

x =

11. Solve

$$2(x + 4) = 3x + 5$$

12. Solve

$$3(2x - 1) = 4x + 7$$

x =

x =

SCORE **5.0** Advanced Content

NC.8.EE.7 Solve equations with fractions or decimals and identify special solutions.

• *I can solve equations with rational coefficients and recognize equations with one, none, or infinitely many solutions.*

Solve. If there is no solution or infinitely many, write that instead.

13. Solve

$$\frac{1}{2}x + 3 = 9$$

x =

14. Solve

$$0.2x + 1 = 3$$

x =

15. Solve

$$\frac{2}{3}x = 8$$

x =

16. Solve

$$3x + 5 = 3x + 5$$

Answer:

17. Solve

$$2x + 4 = 2x - 1$$

Answer:

18. Solve

$$5(x + 2) - 3 = 2(x + 5)$$

x =